

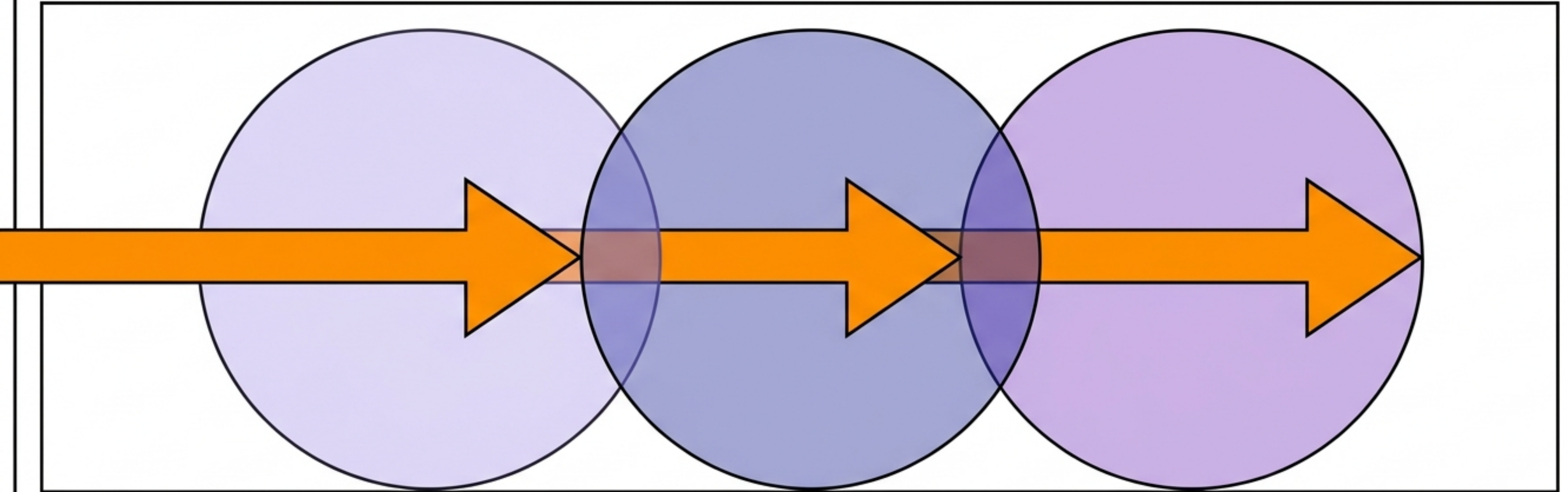
Bitcoin and Quantum Computing

Moving past the "Q-Day" myth:
A strategic framework for assessing
cryptographic risk and network
resilience.

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The Institutional Thesis on Quantum Risk



The Timeline

Not an Imminent Threat.

Quantum computing is a compounding, multi-decade technological evolution, not a sudden, unpredictable event.

The Buffer

The Internet Breaks First.

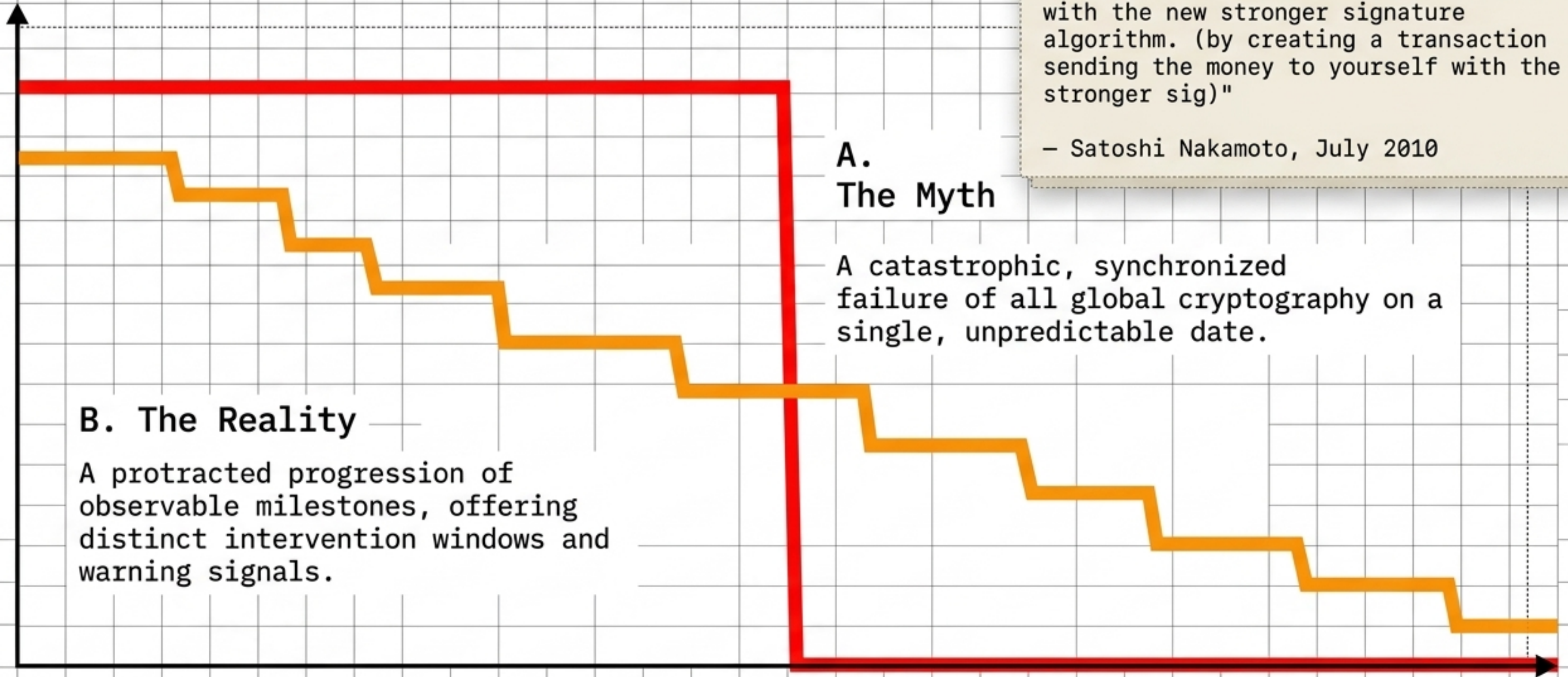
Meaningful quantum breakthroughs will compromise global web infrastructure and banking long before they possess the specific power to attack Bitcoin.

The Economics

High Cost, Slow Yield.

Breaking cryptography will be a protracted, highly expensive thermodynamic effort undertaken at massive cost to the attacker.

Dismantling the “Q-Day” Binary

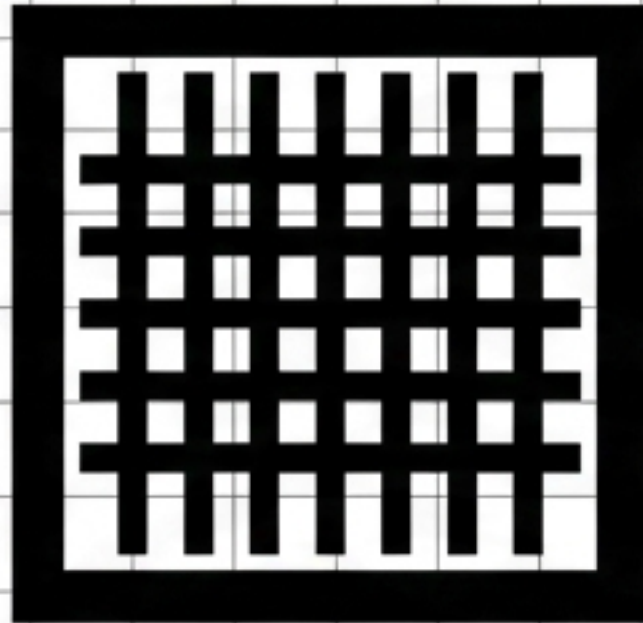


"True, if it happened suddenly. If it happens gradually, we can still transition to something stronger. When you run the upgraded software for the first time, it would re-sign all your money with the new stronger signature algorithm. (by creating a transaction sending the money to yourself with the stronger sig)"

– Satoshi Nakamoto, July 2010

The Cryptographic Target Matrix

Pillar 1: Hash Functions (SHA-256)

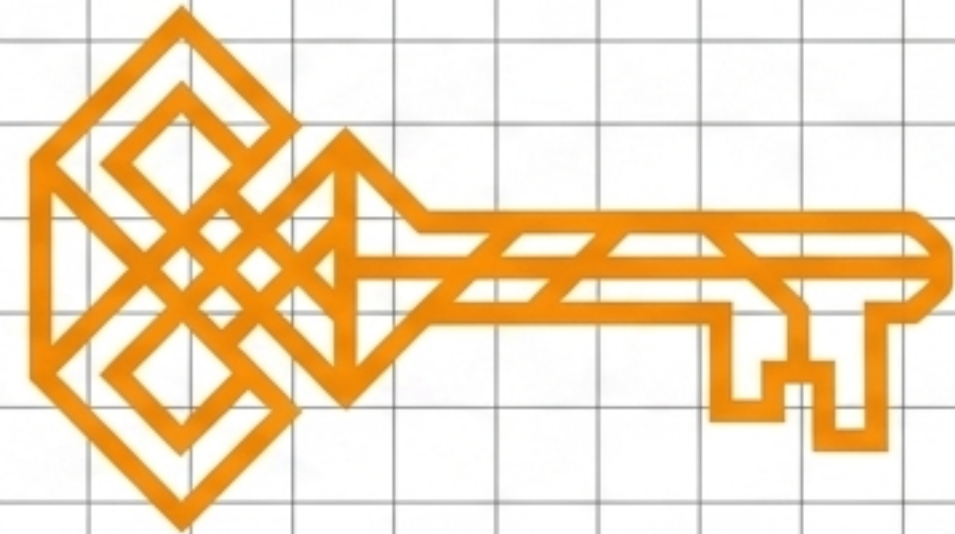


Role: Secures mining, block linking, and transaction ordering.

Quantum Status: **Resistant.**

Current physics research indicates hash functions are highly resistant to quantum algorithms.

Pillar 2: Elliptic Curve Cryptography (secp256k1)



Role: Secures wallet ownership and spending authorization (Digital Signatures).

Quantum Status: **Vulnerable (Eventually).**

Subject to decryption by Shor's algorithm, threatening specific historical address types.

Mapping the Supply Risk Surface

Total Supply Limit: 21,000,000 BTC



8.6%: Vulnerable and Lost (~1.7 Million BTC)

Early P2PK (Pay-to-Public-Key) addresses **widely used pre-2011**. The keys are presumed lost. These are the **prime targets** for future quantum **extraction**.

25.0%: Vulnerable but Migratable (~5 Million BTC)

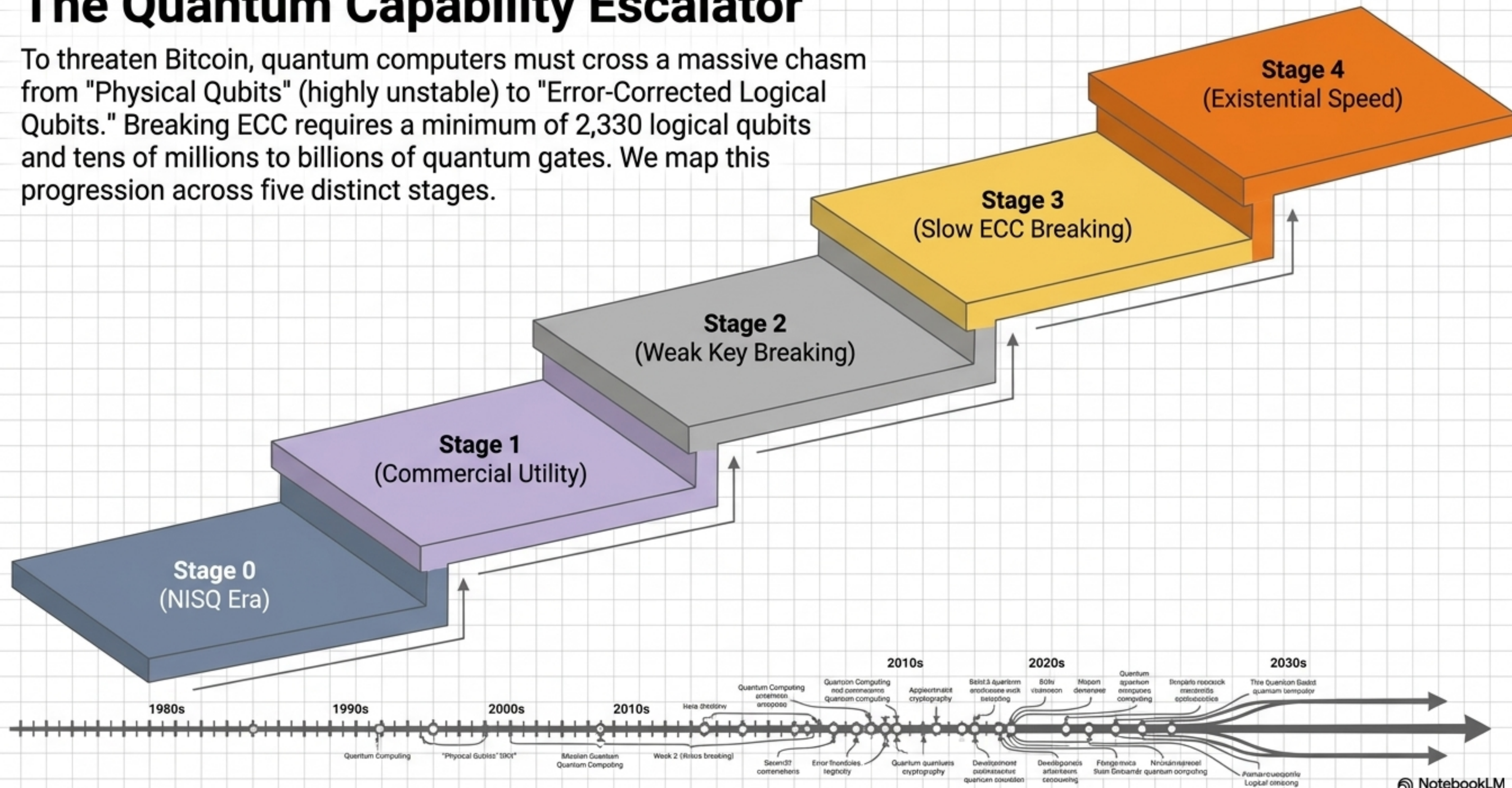
Currently held in reused or P2TR (Taproot) addresses. **At risk in Stage 3**, but owners can move funds to upgraded addresses.

65.4%: Safe (~13 Million BTC)

Held in modern, non-reused, quantum-resistant addresses. **Zero immediate risk**.

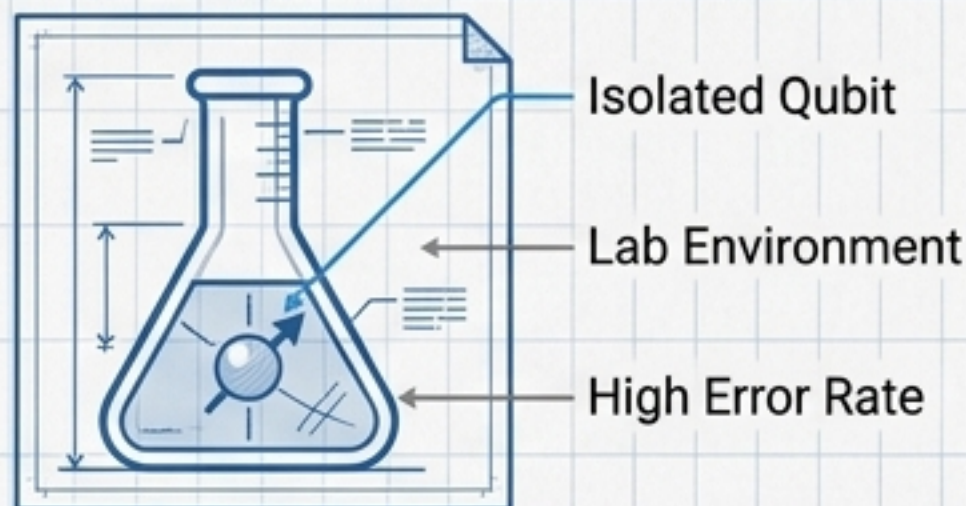
The Quantum Capability Escalator

To threaten Bitcoin, quantum computers must cross a massive chasm from "Physical Qubits" (highly unstable) to "Error-Corrected Logical Qubits." Breaking ECC requires a minimum of 2,330 logical qubits and tens of millions to billions of quantum gates. We map this progression across five distinct stages.



Stages 0 & 1: The NISQ Era to Commercial Advantage

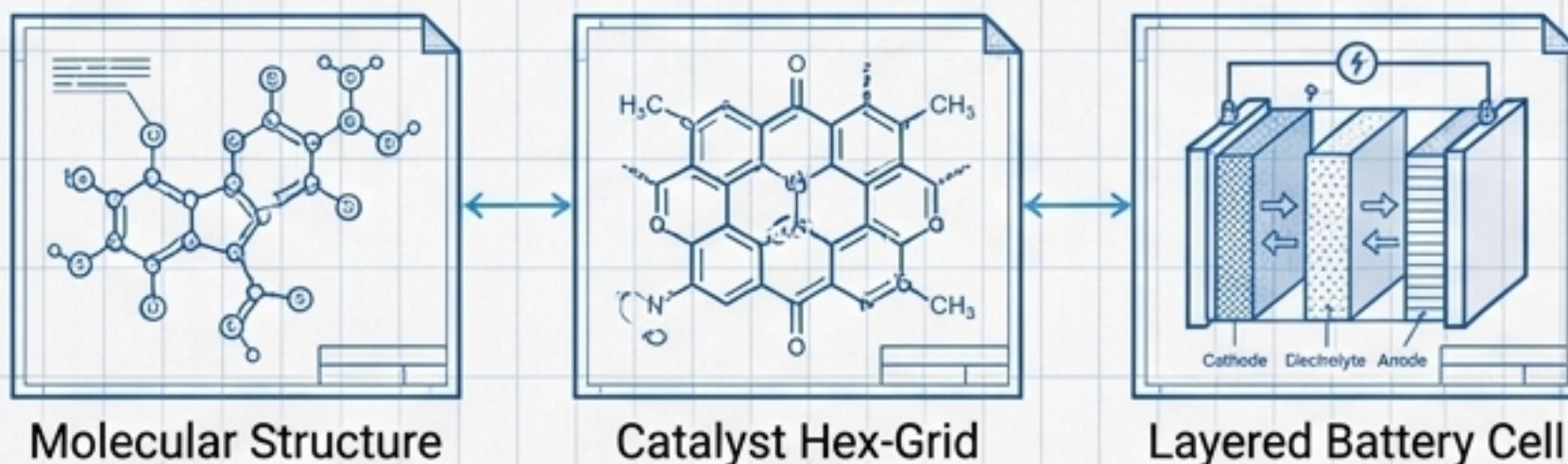
Stage 0: Today (NISQ Era)



~100 logical qubits with high error rates. Systems exist primarily in labs to study quantum physics.

Status: Zero threat to digital assets.

Stage 1: Broad Quantum Advantage



Systems scale to 10^3 – 10^6 circuit depths. They achieve commercial viability by simulating real-world physical systems: homogenous catalyst discovery, fluid dynamics, battery efficiency, and drug binding affinities.

Takeaway: The first profitable quantum companies will revolutionize chemistry and materials science, years before they possess cryptographic relevance.

Stages 2 & 3: The Emergence of the CRQC

Stage 2: Deprecated Vulnerabilities

The first Cryptographically Relevant Quantum Computers (CRQCs) arrive. Attackers target low-hanging fruit: weak keys, older internet protocols, and historical intelligence data ('harvest now, decrypt later').

Stage 3: The Slow Bitcoin Break

A CRQC scales to billions of gates, making 256-bit ECC mathematically solvable—but at an agonizingly slow pace.



The Reality of Theft: A CRQC can only attack one public key at a time. The 1.1 million BTC presumed mined by Satoshi are spread across 22,000 addresses. If a CRQC takes one day to break a key, extracting that sum would take over 60 years.

Stage 4: The 10-Minute Threat Window

The Condition

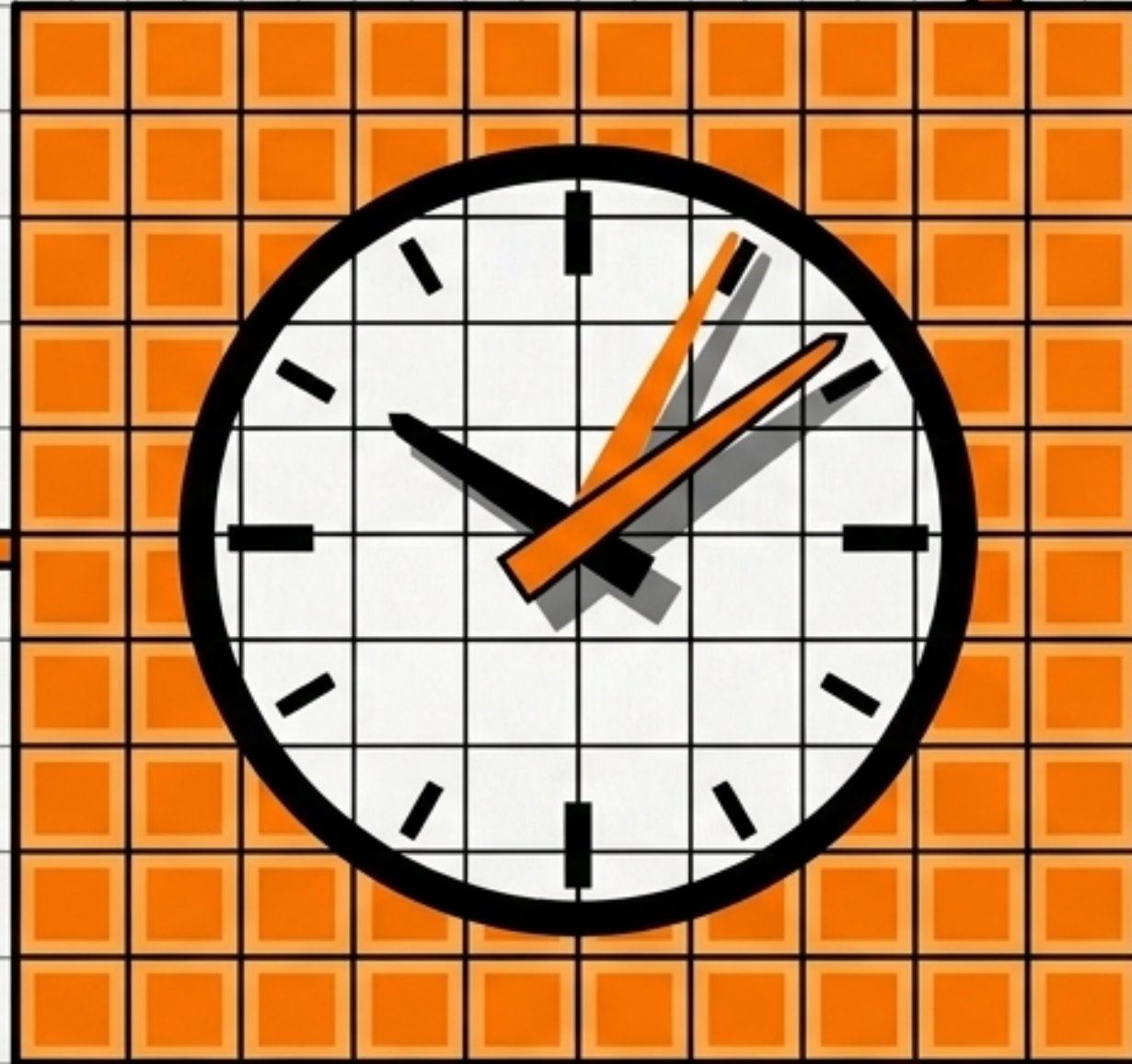
Quantum processing speed accelerates exponentially. A CRQC can now break an ECC public key faster than Bitcoin's 10-minute block interval.

The Impact

The Bitcoin mempool becomes fully exposed. Even users with quantum-resistant addresses become vulnerable the exact moment they attempt to transact, as their public keys are broadcast before the block settles.

The Requirement

Inaction is no longer mathematically possible. Protocol-level, post-quantum cryptographic upgrades are now strictly mandatory for network survival.



The Economics of Attack

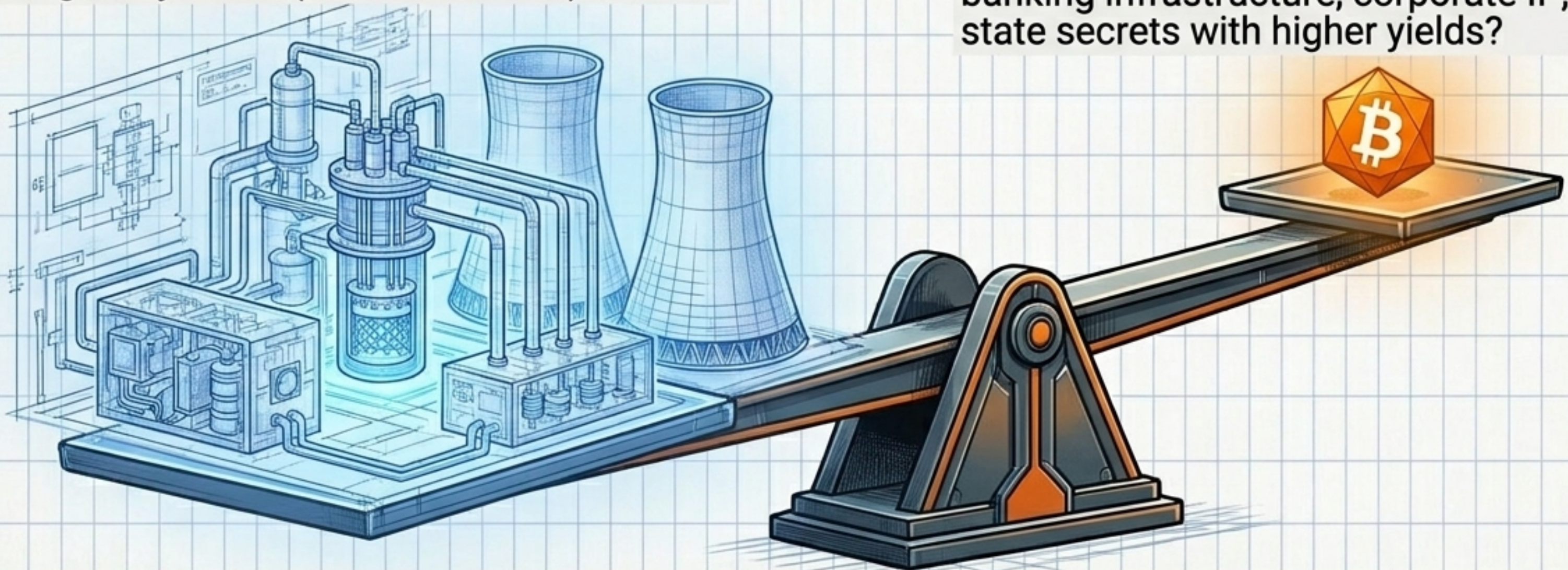
Breaking cryptography is not sending an email; it is a massive thermodynamic undertaking.

Advanced CRQCs require extreme cryogenic cooling and immense operational overhead.

Estimated Cost: ~\$100,000 in electricity alone per single key break (2023 estimates).

Rational Actors

Why spend months and millions cracking a single Satoshi wallet when the same machine could be used to breach global banking infrastructure, corporate IP, or state secrets with higher yields?



Defense Outpaces Offense: PQC is Here

Global Standardization: In 2024, NIST finalized the world's first Post-Quantum Cryptography (PQC) standards, deploying ML-DSA (lattice-based) and SLH-DSA (hash-based) signatures.

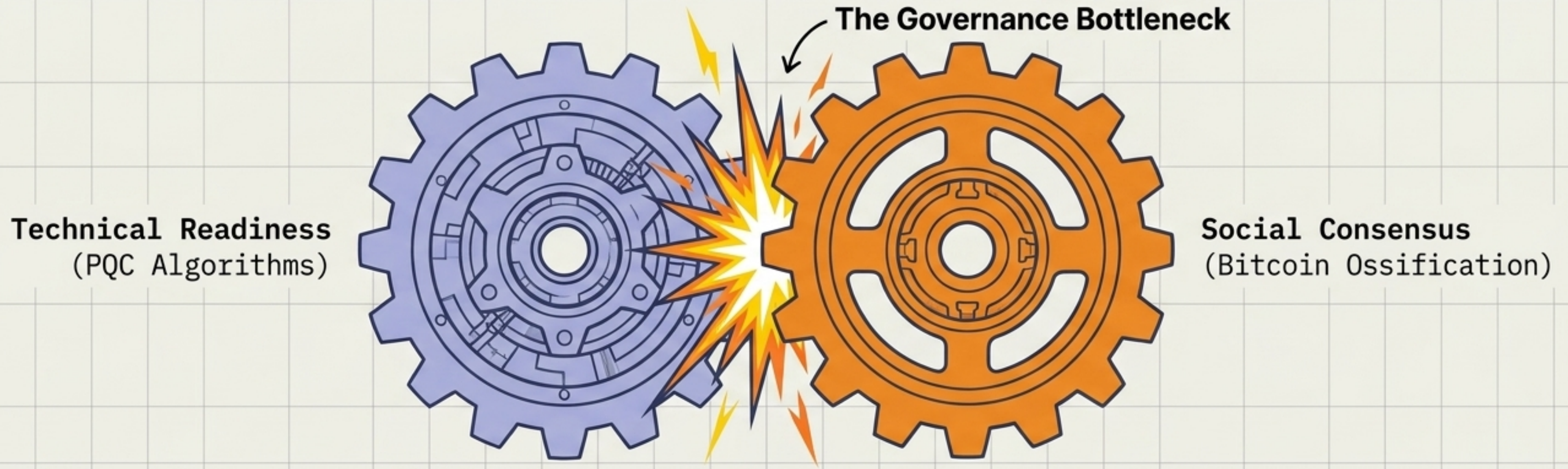
The Internet is Upgrading: Computer security is already transitioning. Browsers, CDNs, and core communication protocols (OpenSSL, OpenSSH) are shipping with PQC by default.

Data Point: Significant portions of modern global web traffic are already moving through quantum-safe channels today, while CRQCs remain decades away.



```
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html
```


The True Bottleneck: Consensus, Not Physics



The Paradox

The math required to defend Bitcoin already exists. The challenge is implementing it on a decentralized network explicitly designed to resist rapid change.

The Engineering Challenge

Bitcoin consensus rules strictly constrain data and compute. PQC signatures are large and complex. Integrating solutions like BIP 360 requires massive, bug-free optimization.

The Social Challenge

Pushing a network-wide soft fork demands overwhelming coordination among decentralized developers, miners, and node operators, risking political division.

The 'Lost Coins' Dilemma

If PQC is successfully implemented, what happens to the **1.7M vulnerable BTC (P2PK)** whose owners are dead or lost?



Path A: Burn & Freeze (Censorship)

- The network initiates a consensus change to permanently lock quantum-vulnerable coins.
- **Pro:** Prevents market supply shock.
- **Con:** Breaks Bitcoin's core ethos of censorship resistance ("confiscation by fiat").

Path B: Do Nothing (Sovereignty)

- The network refuses to interfere. Capable quantum attackers slowly mine these abandoned coins over decades.
- **Pro:** Upholds pure decentralization.
- **Con:** Allows attackers to extract value and introduces gradual sell pressure.

Strategic Foresight: Three Plausible Scenarios

	Optimistic (Quantum Winter)	Balanced (Most Likely / 10-20 Years)	Pessimistic (AI-Driven Leap)
Technology		Commercial quantum succeeds, creating urgency.	Sudden, unexpected CRQC emergence. Global infrastructure panics.
Bitcoin Network Response		The community engages in a contentious but ultimately successful PQC soft fork. Network secured well before Stage 3.	Bitcoin forces an emergency, chaotic fork. Survival due to underlying thermodynamic proof-of-work, but extreme volatility.
Market Impact		Temporary turbulence, ultimate resilience.	High friction and community splits.

Strategic Posture for the Allocator

1. Monitor Milestones, Not Dates

Ignore apocalyptic “Q-Day” predictions. Track the verified scaling of error-corrected logical qubits and the success of commercial quantum chemistry applications.

2. Audit Custody Architectures

As PQC standards finalize, ensure institutional holdings are actively migrating away from legacy address structures toward modernized, quantum-resistant wallet architectures.

3. Embrace the Friction

Recognize that Bitcoin’s notoriously slow, contentious upgrade cycle is not a bug; it is its greatest feature. This ossification acts as the ultimate safeguard against introducing systemic vulnerabilities via rushed code.